WORKED SOLUTION

STEP III 2020, Question 12

Probability and Statistics • Geometric Distribution Independence

STEP III 2020, Question 12

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

Everything is a function of the pair (X, Y) of independent geometric variables, with $\mathrm{P}(X = x,\, Y = y) = p^2 q^{x+y-2}$. Sums and differences are best handled by adding up this joint probability along a line in the (x, y) grid, while maxima and minima are easiest through cumulative probabilities such as $\mathrm{P}(T \le t)$. For independence, look for a value of one variable that pins down the other completely.

Solution

Since A tosses until the first head, X takes the value $k \ge 1$ when the first $k-1$ tosses are tails and the kth is a head:

$$\mathrm{P}(X = k) = pq^{k-1}, \qquad k = 1, 2, 3, \ldots,$$

and the same holds for Y. The two players toss separately, so X and Y are independent, and

$$\mathrm{P}(X = x,\, Y = y) = p^2 q^{x+y-2} \qquad (x, y \ge 1).$$

Two facts we use repeatedly: $\mathrm{P}(X > k) = q^k$ (the first k tosses are all tails), so $\mathrm{P}(X \le k) = 1 - q^k$; and $1 - q^2 = (1-q)(1+q) = p(1+q)$.

Part (i)

The sum. S takes values $s \ge 2$. For $S = s$ we need $X = x$ and $Y = s - x$ with $1 \le x \le s-1$. Each of these $s-1$ outcomes has probability $p^2 q^{s-2}$, so

$$\mathrm{P}(S = s) = (s-1)p^2 q^{s-2}, \qquad s = 2, 3, 4, \ldots$$

The maximum. $T \le t$ exactly when both $X \le t$ and $Y \le t$, so by independence

$\mathrm{P}(T \le t) = (1-q^t)^2$. Hence, for $t \ge 1$,

$$\begin{aligned}\mathrm{P}(T = t) &= \mathrm{P}(T \le t) - \mathrm{P}(T \le t-1) = (1-q^t)^2 - (1-q^{t-1})^2 \\ &= [(1-q^t) - (1-q^{t-1})][(1-q^t) + (1-q^{t-1})] \\ &= (q^{t-1} - q^t)(2 - q^{t-1} - q^t) \\ &= pq^{t-1}(2 - q^{t-1} - q^t),\end{aligned}$$

using $q^{t-1} - q^t = q^{t-1}(1-q) = pq^{t-1}$. This is the required result. (For $t = 1$ the formula gives $p(1-q) = p^2$, correctly.)

Part (ii)

The difference. For $U = 0$ we need $X = Y = k$ for some $k \ge 1$:

$$\mathrm{P}(U = 0) = \sum_{k=1}^{\infty} p^2 q^{2k-2} = \frac{p^2}{1-q^2} = \frac{p^2}{p(1+q)} = \frac{p}{1+q}.$$

For $u \ge 1$, either $Y = X + u$ or $X = Y + u$; these are disjoint and by symmetry equally likely, so

$$\mathrm{P}(U = u) = 2\sum_{k=1}^{\infty} \mathrm{P}(X = k,\, Y = k+u) = 2\sum_{k=1}^{\infty} p^2 q^{2k+u-2} = \frac{2p^2 q^u}{1-q^2} = \frac{2pq^u}{1+q}.$$

The minimum. $W > w$ exactly when both $X > w$ and $Y > w$, so $\mathrm{P}(W > w) = q^{2w}$ and, for $w \ge 1$,

$$\mathrm{P}(W = w) = \mathrm{P}(W > w-1) - \mathrm{P}(W > w) = q^{2w-2} - q^{2w} = (1-q^2)\, q^{2(w-1)}.$$

So W is itself geometric, with "success" probability $1 - q^2$: each round of one toss by A and one by B produces at least one head with probability $1 - q^2$.

Part (iii)

If $S = 2$ then, since $X, Y \ge 1$, we must have $X = Y = 1$, and then $T = 1$. So $S = 2$ and $T = 3$ cannot happen together:

$$\mathrm{P}(S = 2 \text{ and } T = 3) = 0.$$

On the other hand $\mathrm{P}(S = 2) = p^2 > 0$ and $\mathrm{P}(T = 3) = pq^2(2 - q^2 - q^3) > 0$, because $0 < q < 1$ gives $q^2 + q^3 < 2$. Hence $\mathrm{P}(S = 2) \times \mathrm{P}(T = 3) > 0$, and the two sides differ.

Part (iv)

U and W are independent. We must show $\mathrm{P}(U = u,\, W = w) = \mathrm{P}(U = u)\,\mathrm{P}(W = w)$ for every $u \ge 0$ and $w \ge 1$. Knowing both U and W tells us the smaller value is w and the larger is $w + u$.

Ask yourself Which pairs (X, Y) give $U = u$ and $W = w$?

If $u = 0$ the only pair is (w, w). If $u \geq 1$ there are exactly two: $(w,\ w+u)$ and $(w+u,\ w)$, each with probability p^2q^{2w+u-2}.

For $u = 0$, $\mathrm{P}(U = 0,\ W = w) = p^2q^{2w-2}$, while

$$\mathrm{P}(U = 0)\,\mathrm{P}(W = w) = \frac{p}{1+q} \cdot p(1+q)\,q^{2w-2} = p^2q^{2w-2}.$$

For $u \geq 1$, $\mathrm{P}(U = u,\ W = w) = 2p^2q^{2w+u-2}$, while

$$\mathrm{P}(U = u)\,\mathrm{P}(W = w) = \frac{2pq^u}{1+q} \cdot p(1+q)\,q^{2w-2} = 2p^2q^{2w+u-2}.$$

These agree in every case, so U and W are independent.

No other pair is independent. The remaining pairs are (S, T), (S, U), (S, W), (T, U) and (T, W). To show two variables are *not* independent, one pair of values with $\mathrm{P}(A = a,\ B = b) \neq \mathrm{P}(A = a)\,\mathrm{P}(B = b)$ is enough.

Ask yourself Which single value of S or T forces everything else?

The event $S = 2$ and the event $T = 1$ are the same event, $\{X = Y = 1\}$, of probability $p^2 > 0$; on it $T = 1$, $U = 0$ and $W = 1$. Every value of T, U and W in the list below has positive probability by the formulas in parts (i) and (ii), since $0 < p, q < 1$. So:

- (S, T): $\mathrm{P}(S = 2,\ T = 3) = 0 \neq \mathrm{P}(S = 2)\,\mathrm{P}(T = 3)$, by part (iii);
- (S, U): $\mathrm{P}(S = 2,\ U = 1) = 0$, but $\mathrm{P}(S = 2)\,\mathrm{P}(U = 1) > 0$;
- (S, W): $\mathrm{P}(S = 2,\ W = 2) = 0$, but $\mathrm{P}(S = 2)\,\mathrm{P}(W = 2) > 0$;
- (T, U): $\mathrm{P}(T = 1,\ U = 1) = 0$, but $\mathrm{P}(T = 1)\,\mathrm{P}(U = 1) > 0$;
- (T, W): $\mathrm{P}(T = 1,\ W = 2) = 0$, but $\mathrm{P}(T = 1)\,\mathrm{P}(W = 2) > 0$.

So U and W form the only independent pair among S, T, U, W.

Checks

- Each distribution sums to 1: for instance $\sum_{t\geq 1} \mathrm{P}(T = t)$ telescopes to $\lim(1 - q^t)^2 = 1$, and $\frac{p}{1+q} + \sum_{u\geq 1} \frac{2pq^u}{1+q} = \frac{p}{1+q}\left(1 + \frac{2q}{p}\right) = \frac{p+2q}{1+q} = 1$.
- With $p = q = \frac{1}{2}$: $\mathrm{P}(T = 2) = \frac{1}{4}\left(2 - \frac{1}{2} - \frac{1}{4}\right) = \frac{5}{16}$, which matches listing the three outcomes $(1, 2), (2, 1), (2, 2)$, with probabilities $\frac{1}{8} + \frac{1}{8} + \frac{1}{16}$.
- With $p = q = \frac{1}{2}$, $\mathrm{P}(U = 1,\ W = 1) = \mathrm{P}\{(1, 2), (2, 1)\} = \frac{1}{4}$, and $\mathrm{P}(U = 1)\,\mathrm{P}(W = 1) = \frac{1}{3} \times \frac{3}{4} = \frac{1}{4}$, as independence requires.
- All the formulas, and the independence of U and W, were confirmed by computer by summing the joint distribution for $p = 0.3$, 0.55 and 0.8.

Common slips

- Forgetting the factor 2 in $\mathrm{P}(U = u)$ for $u \geq 1$, or wrongly including it for $u = 0$, where $X = Y$ counts once.
- Trying to find $\mathrm{P}(T = t)$ by adding cases directly and double-counting $X = Y = t$; the cumulative approach avoids this.
- Thinking that checking a single pair of values proves independence: that needs *all* u and w, whereas one failure is enough for dependence.

Beyond STEP

The independence of U and W is the discrete form of the memoryless property: once the first head appears at time W, the other player's remaining wait is a fresh geometric variable, regardless of how long it took. The continuous analogue, for independent exponential variables, says the minimum and the gap to the maximum are independent, which underlies Poisson processes and queueing theory. Grimmett and Welsh, *Probability: An Introduction*, develops these ideas.

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